

$P(v, t | v_0, t_0) dv$: probability finding the Brownian particles have velocity $[v, v+dv]$ given that the particle is at v_0 at time t_0

Consider stationary process, then

$$P(v, t | v_0, t_0) = P(v, t - t_0 | v_0, 0)$$

i.e. we consider that v_0 is the value at time $t=0$ and write $P(v, t | v_0)$

$$\int_{-\infty}^{\infty} P(v, t | v_0) dv = 1$$

$$\lim_{t \rightarrow 0} P(v, t | v_0) = \delta(v - v_0)$$

(initial condition)

for $t \rightarrow \infty$ system forget its history v_0

velocity should follow Maxwell distribution

$$\lim_{t \rightarrow +\infty} P(v, t | v_0) \propto e^{-\beta(\frac{1}{2}mv^2)}$$

$$\text{with proper normalization} = \left(\frac{m\beta}{2\pi}\right)^{\frac{1}{2}} e^{-\beta(\frac{1}{2}mv^2)}$$

Associated Langevin Eq is

$$m \frac{dv}{dt} = -m\gamma v + R(t)$$

F. Reif Chp 15
R. Zwanzig Chp 1 & 2

Kramers-Moyal expansion

$$\frac{\partial P(v, t | v_0)}{\partial t} dv \cdot \tau$$

rate of

increase in prob
finding particle

in $[v, v+dv]$

$$\approx [P(v, t+\tau | v_0) - P(v, t | v_0)] dv$$

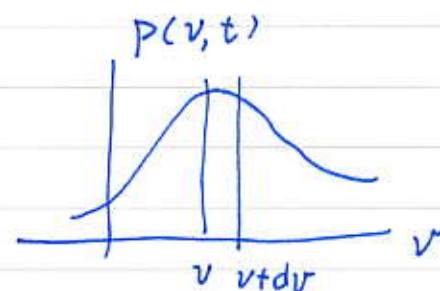
$$= - \int_{v'} P(v, t | v_0) P(v', \tau | v) dv' +$$

↑
after time elapsed by τ
the prob. that particle has
velocity v' is $P(v', \tau | v) dv'$

$$P(A) P(B|A) = P(A \cap B)$$

$$+ \int_{v'} P(v', t | v_0) dv' P(v, \tau | v')$$

another velocity v' can be come the
current one.



$$\text{but } \int_{v'} P(v', \tau | v) dv' = 1$$

$$v' = v - \xi \leftarrow \text{small change}$$

(see derivation
by Reif)

$$\frac{\partial P(v, t | v_0)}{\partial t} \tau = - P(v, t | v_0) +$$

$$\int_{-\infty}^{\infty} P(v - \xi, t | v_0) P(v, \tau | v - \xi) d\xi$$

$\tau \rightarrow 0$. ξ small. P change slowly with respect to v $x_0 = v$

$$P(v - \xi, t | v_0) P(v, \tau | v - \xi)$$

$$x = v - \xi = x_0 - \xi$$

$$= P(v - \xi, t | v_0) P(v - \xi + \xi, \tau | v - \xi) = P(x, t | v_0) P(x + \xi, \tau | x)$$

$$= P(x_0, t | v_0) P(x_0 + \xi, \tau | x_0) + \frac{\partial}{\partial x} P(x, t | v_0) P(x + \xi, \tau | x) \Big|_{x=x_0}$$

$$= P(v, t | v_0) P(v + \xi, \tau | v) + (-\xi) \frac{\partial}{\partial v} (P(v, t | v_0) P(v + \xi, \tau | v)) \\ + \frac{1}{2} (-\xi)^2 \frac{\partial^2}{\partial v^2} P(v, t | v_0) P(v + \xi, \tau | v) + \dots$$

$$\frac{\partial P(v, t | v_0)}{\partial t} \tau = - P(v, t | v_0) + \int_{-\infty}^{\infty} P(v, t | v_0) P(v + \xi, \tau | v) d\xi$$

↓ cancel

$$+ \frac{\partial}{\partial v} P(v, t | v_0) \int_{-\infty}^{\infty} (-\xi) P(v + \xi, \tau | v) d\xi$$

$$+ \frac{1}{2} \frac{\partial^2}{\partial v^2} P(v, t | v_0) \int_{-\infty}^{\infty} \xi^2 P(v + \xi, \tau | v) d\xi + \dots$$

$v + \xi = v' = v(\tau)$

$$M_n = \frac{1}{\tau} \int_{-\infty}^{\infty} \xi^n P(v + \xi, \tau | v) d\xi = \frac{1}{\tau} \langle [v(\tau) - v]^n \rangle$$

change of velocity after time τ

$$\frac{\partial P}{\partial t} = - \frac{\partial}{\partial v} (M_1 P) + \frac{1}{2} \frac{\partial^2}{\partial v^2} (M_2 P) \leftarrow \text{Fokker-Planck Eq.}$$

$$M_1 = \frac{1}{\tau} \langle (v(\tau) - v) \rangle = \frac{1}{\tau} [\langle v(\tau) \rangle - v] \quad v(\tau=0) = v$$

average velocity

$$\langle v(\tau) \rangle = v e^{-\gamma \tau} = v [1 - \gamma \tau + O(\tau^2)]$$

$$= v - \gamma v \tau + \dots$$

$$= -\gamma v$$

$$M_2 = \frac{1}{\tau} \langle (v(\tau) - v)^2 \rangle = \frac{2k_B T}{m} \gamma = 2D\gamma^2$$

$$M_n = 0 \quad \text{for } n \geq 3 \quad \text{for gaussian process!}$$

$$v(t) = v(0) e^{-\gamma t} + \int_0^t \frac{R(\tau)}{m} e^{-\gamma(t-\tau)} d\tau$$

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t small

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$$\begin{aligned} \lim_{t \rightarrow \infty} \frac{1}{t} \langle (v(t) - v(0))^2 \rangle &= \left\langle \left[v(0) (e^{-\gamma t} - 1) + \int_0^t \frac{R(\tau)}{m} e^{-\gamma(t-\tau)} d\tau \right]^2 \right\rangle \\ &= \frac{1}{t} \left[\langle (v(0) [-\gamma t + \dots])^2 \rangle + \langle R(\tau) R(\tau') \rangle \frac{1}{m^2} \right] \\ &= \frac{1}{t} \frac{C \cdot t}{m^2} \quad C = 2m\gamma k_B T \end{aligned}$$

at $t \rightarrow \infty$ No cross term $\int_0^t \int_0^t$

→ with spectrum M_1, M_2
we have

$$\frac{\partial P}{\partial t} = \gamma \frac{\partial}{\partial v} (vP) + \underbrace{\gamma \frac{k_B T}{m}}_{\gamma D} \frac{\partial^2 P}{\partial v^2}$$

$$D = \frac{k_B T}{m\gamma}$$

solution of Fokker-Planck equation

Let $\frac{\partial^2 P}{\partial v^2} = 0$ first

follow F.R.P.f

$$\frac{\partial P}{\partial t} - \gamma v \frac{\partial P}{\partial v} = \gamma P$$

multiply by $\lambda(v, t)$ so that

$$\lambda \frac{\partial P}{\partial t} - \lambda \gamma v \frac{\partial P}{\partial v} = \lambda \gamma P \quad \text{is a total differential}$$

$$\frac{\partial P}{\partial t} dt + \frac{\partial P}{\partial v} dv = dP$$

we identify

$$\left. \begin{aligned} dt &= \lambda \\ dv &= -\lambda \gamma v \\ dP &= \lambda \gamma P \end{aligned} \right\}$$

$$\frac{dv}{dt} = -\gamma v$$

$$\frac{dP}{dt} = \gamma P$$

$$= \left[\frac{m}{2\pi k_B T (1 - e^{-2\gamma t})} \right]^{\frac{1}{2}} \exp \left[- \frac{m (v - v_0 e^{-\gamma t})^2}{2 k_B T (1 - e^{-2\gamma t})} \right]$$

$t \rightarrow \infty$ reduce to Maxwell distribution

$$\frac{\partial P}{\partial t} = \gamma \frac{\partial}{\partial v} (v P) + \frac{\gamma k_B T}{m} \frac{\partial^2 P}{\partial v^2}$$

equilibrium distribution 1st $t \rightarrow \infty$ } doesn't change with time

$$\frac{\partial}{\partial v} (v P) + \frac{k_B T}{m} \frac{\partial^2 P}{\partial v^2} = 0$$

$$\text{or } v P + \frac{k_B T}{m} \frac{\partial P}{\partial v} = \text{const.}$$

const must equal 0. b/c $\int_{-\infty}^{+\infty} P dv = 1$

integrate on both side

$$\left| \int_{-\infty}^{+\infty} v P dv \right| < M \text{ bounded}$$

$$\int_{-\infty}^{+\infty} \text{const} dv \text{ must be finite}$$

$$- \frac{m v dv}{k_B T} = \frac{dP}{P}$$

$$\rightarrow P = P_0 e^{-\beta \left(\frac{1}{2} m v^2 \right)} \quad \beta = \frac{1}{k_B T}$$

← Maxwell velocity distribution

From Langevin to Fokker-Planck, general setting

$$\dot{\vec{X}} = G(\vec{X}) + g(\vec{X}) \xi$$

$\vec{X}$ vector $\rightarrow$ G matrix

$$\langle \xi \xi^T \rangle = 2D \delta(t-t')$$

D is const

$$\frac{\partial P}{\partial t} = - \frac{\partial}{\partial \vec{X}}^T G P + \left(\frac{\partial}{\partial \vec{X}} \right)^T g D \left[g^T \frac{\partial^2}{\partial \vec{X}^2} P \right]$$

$\frac{\partial}{\partial \vec{X}}$ row vector

$$\vec{X} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{pmatrix} \text{ is arbitrary vector}$$

when gas can be treated classically

$$S = Nk_B \left\{ \ln \left(\frac{V}{N \lambda^3} \right) + \frac{5}{2} \right\}$$

$$\lambda = \frac{h}{\sqrt{2m k_B T}}$$

$$\left(\frac{V}{N} \right)^{1/3} = \frac{\lambda}{b} \leftarrow \text{size of particle}$$

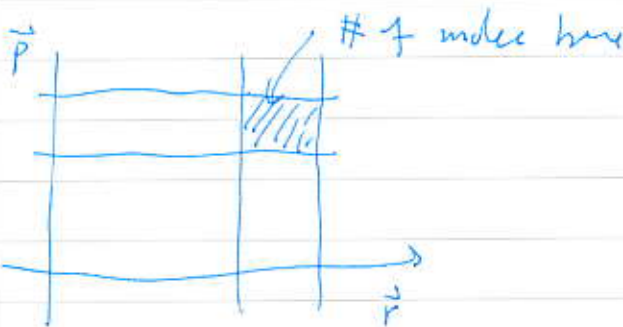
$$\lambda = \frac{10^{-34}}{\sqrt{10^{-26} \times 1.38 \times 10^{-23} \times 300 \text{ K}}}$$

$$= 1.5 \times 10^{-11} \text{ m} = 0.15 \text{ \AA}$$

$$b \approx 10^{-30} \text{ \AA}$$

$\lambda \gg b$

$f(\vec{r}, \vec{p}, t) d^3r d^3p = \# \text{ of molecules having position } \vec{r} \text{ in volume } d^3r$
or $\text{momentum } \vec{p} \text{ in volume } d^3p$



joint distribution makes sense only in classical regime

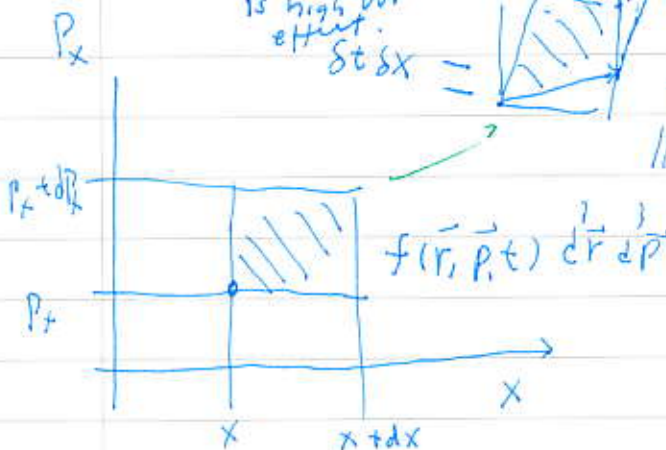
$$\int f(\vec{r}, \vec{p}, t) d^3r d^3p = N$$

change in volume is high order effect.

$$\delta t \delta x =$$



volume is preserved, provided a single particle system is Hamiltonian.



$$f(\vec{r} + \delta \vec{r}, \vec{p} + \delta \vec{p}, t + \delta t) d^3r d^3p$$

volume do not change (proof of Liouville theorem)

external force only

$$\frac{Df}{Dt} = \left(\frac{\partial f}{\partial t} + \frac{\vec{p}}{m} \cdot \vec{\nabla}_r + \vec{F} \cdot \vec{\nabla}_p \right) f = \left(\frac{\partial f}{\partial t} \right)_{\text{coll}}$$

$$f(\vec{r}, \vec{p}) = N \int p(\vec{r}, \vec{p}; \vec{r}_1, \vec{p}_1, \vec{r}_2, \vec{p}_2, \dots) d\vec{r}_1 d\vec{p}_1 \dots d\vec{r}_N d\vec{p}_N$$

$$\sum_i \int p(\dots)$$

sum of 1st particle in $d\vec{r} d\vec{p}$, 2nd in $d\vec{r} d\vec{p}$, etc.

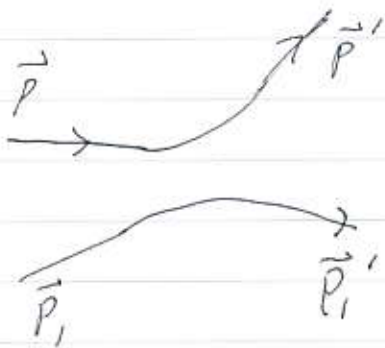
satisfying

$$\frac{Df}{Dt} = \left(\frac{\partial f}{\partial t} + \vec{v} \cdot \vec{\nabla}_r + \vec{F} \cdot \vec{\nabla}_p \right) f = \left(\frac{\partial f}{\partial t} \right)_{\text{coli}}$$

$\bar{R} = R - R$ ~~is not~~ $\bar{R}$ # of collisions occurring ~~per unit time~~ ^{Date} in which one of the initial molecules

$R \delta t d\vec{r} d\vec{p}$ = no of collisions occurring during the time between t and $t + \delta t$ in which one of the initial molecules is in $d\vec{r} d\vec{p}$ at $(\vec{r}, \vec{p})$

$\bar{R} \delta t d\vec{r} d\vec{p}$ = no. of collisions occurring during the time between t and $t + \delta t$ in which one of the final molecules is in $d\vec{r} d\vec{p}$ at $(\vec{r}, \vec{p})$.

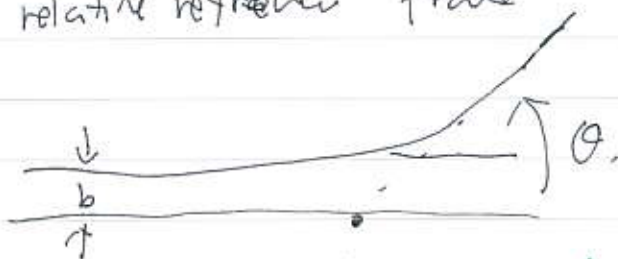


$$\vec{p} + \vec{p}_1 = \vec{p}' + \vec{p}_1'$$

$$\epsilon(\vec{p}) + \epsilon(\vec{p}_1) = \epsilon(\vec{p}') + \epsilon(\vec{p}_1')$$

$$\epsilon = \frac{p^2}{2m}$$

in relative reference frame



given $\vec{p}, \vec{p}_1$

b, ϕ or (θ, ϕ) can still vary.

target correct molecule with $\vec{p}$ is now target

$$(p, p_1 | \sigma | p', p'_1) \propto \delta(p + p_1 - p' - p'_1) \delta[\epsilon(p) + \epsilon(p_1) - \epsilon(p') - \epsilon(p'_1)]$$

differential cross-section
target fixed, in coming flux is I
 $I \sigma(\theta, \phi) d\Omega$ = # of incident molecules scattered per second into solid angle $d\Omega$ in direction (θ, ϕ)

$$\left(\frac{\partial f}{\partial t}\right)_{\text{coll}} = - \int d\vec{p}_1 \int d\vec{p}_1' \int d\vec{p}' \sigma(p, p_1 | p' p_1') (f f_1 - f' f_1')$$

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1st is 2nd is

R

R

$$f_1 = f(r, p_1, t)$$

$$f' = f(r, p', t)$$

$$f_1' = f(r, p_1', t)$$

another form

$$= - \int d\vec{p}_1 d\Omega |v_1 - v| \sigma(\theta, \phi) (f f_1 - f' f_1')$$

= 4X3

$\vec{p}_1, \vec{p}_1', \vec{p}, \vec{p}'$ have 12 degrees of freedom
conservation law for p & ϵ take away 4 degree of

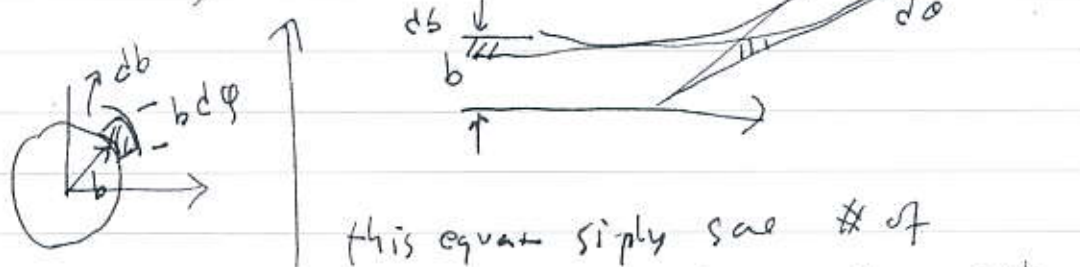
still left with 8 degrees of freedom which is
 $\vec{p}, \vec{p}_1, d\Omega \leftarrow$ out going angles in relative
reference frame.
3 + 3 + 2

Boltzmann

b: impact parameter: we can think of this as precisely define σ .

Important relation between σ and impact parameter b

$$d\omega = b db d\phi = \sigma d\Omega$$



this equation simply says # of particles come from the area $db \cdot b d\phi$ must go out to solid angle $d\Omega$, which is $\sigma d\Omega$ by definition

4 April 07

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Two-body scattering problem

$$H(\vec{r}_1, \vec{r}_2, \vec{p}_1, \vec{p}_2) = \frac{p_1^2}{2m_1} + \frac{p_2^2}{2m_2} + V(|\vec{r}_1 - \vec{r}_2|)$$

reduce to one-body problem

$$\begin{cases} \vec{r} = \vec{r}_2 - \vec{r}_1 & \text{relative position} \\ \vec{R} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2} & \text{center of mass} \end{cases}$$

moment of the center of mass

$$\vec{P} = M \dot{\vec{R}} = (m_1 + m_2) \dot{\vec{R}}$$

$$\mu = \frac{m_1 m_2}{m_1 + m_2}, \quad \frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2}$$

solve $\vec{r}_1$ & $\vec{r}_2$ in terms $\vec{r}$ & $\vec{R}$

$$\vec{R} = \frac{m_1 \vec{r}_1 + m_2 (\vec{r} + \vec{r}_1)}{m_1 + m_2} \quad (m_1 + m_2) \vec{R} = (m_1 + m_2) \vec{r}_1 + m_2 \vec{r}$$

$$\vec{r}_1 = \vec{R} - \frac{m_2}{m_1 + m_2} \vec{r}, \quad \vec{r}_2 = \vec{r} + \vec{r}_1 = \vec{R} + \frac{m_1}{m_1 + m_2} \vec{r}$$

$$\vec{p}_1 = m_1 \dot{\vec{r}}_1 = m_1 \dot{\vec{R}} - \frac{m_1 m_2}{m_1 + m_2} \dot{\vec{r}} = m_1 \dot{\vec{R}} - \mu \dot{\vec{r}}$$

$$\vec{p}_2 = m_2 \dot{\vec{r}}_2 = m_2 \dot{\vec{R}} + \frac{m_1 m_2}{m_1 + m_2} \dot{\vec{r}} = m_2 \dot{\vec{R}} + \mu \dot{\vec{r}}$$

$$\vec{p}_1 + \vec{p}_2 = \vec{P} = \text{const.} \quad \dot{\vec{r}} = \vec{v}$$

$$\frac{\vec{p}_1^2}{2m_1} = \frac{1}{2m_1} (m_1^2 \dot{\vec{R}}^2 + \mu^2 \dot{\vec{r}}^2 - 2m_1 \mu \dot{\vec{R}} \cdot \dot{\vec{r}})$$

$$\frac{\vec{p}_2^2}{2m_2} = \frac{1}{2m_2} (m_2^2 \dot{\vec{R}}^2 + \mu^2 \dot{\vec{r}}^2 + 2m_2 \mu \dot{\vec{R}} \cdot \dot{\vec{r}})$$

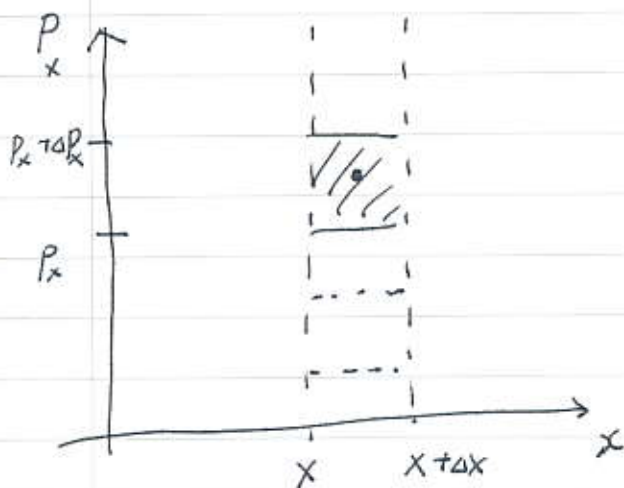
$$= \frac{1}{2} (m_1 + m_2) \dot{\vec{R}}^2 + \frac{\mu^2}{2(m_1 + m_2)} \left(\frac{1}{m_1} + \frac{1}{m_2} \right) v^2$$

$$= \frac{1}{2M} \dot{\vec{R}}^2 + \frac{1}{2} [\mu] \dot{\vec{r}}^2$$

shown.

The collision term in Boltzmann Eq.

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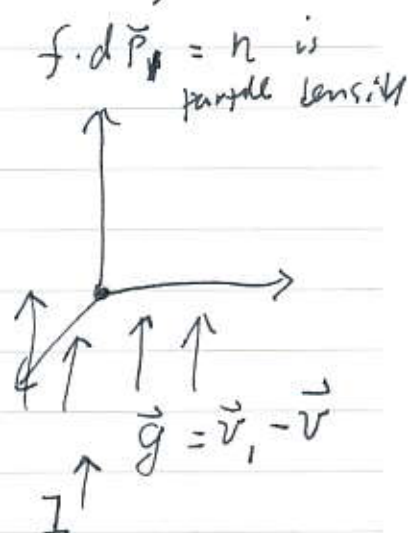
sizes of molecules are much smaller than Δx each $d\vec{r}$ still has a lot of molecules $(\sigma_{tot} \ll \Delta x^2)$

molecules inside $d\vec{r} d\vec{p}$ do not collide as they have the same velocity

consider One molecule in $d\vec{r} d\vec{p}$ and all the molecules in $d\vec{r} d\vec{p}_1$ there is the flux of

$$I = f(\vec{r}, \vec{p}, t) |\vec{v}_1 - \vec{v}| \cdot d\vec{p}_1$$

the # of particles collided with the one (per second) is



$$I d\sigma = I \sigma(\theta, \phi) d\Omega = f(\vec{r}, \vec{p}_1, t) \sigma(\theta, \phi) d\Omega d\vec{p}_1$$

since there are $f(\vec{r}, \vec{p}) d\vec{p}$ molecules there

The total collision per second is (in unit $d\vec{r}$ and $d\vec{p}$)

$$I \sigma \cdot f(\vec{r}, \vec{p}) = f f_1 \sigma |\vec{v}_1 - \vec{v}| d\Omega d\vec{p}_1$$

sum over all possible $\vec{p}_1$ and outcome $d\Omega$.

$$\left(\frac{\partial f}{\partial t} \right)_{\text{col}} = \# \text{ of collision per unit time in unit space and momenta volumes.}$$

Assignment 5

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$$G = G_0 + a(T-T^*)M^2 - \frac{1}{3}cM^3 + \frac{1}{4}dM^4$$

← cubic eq.

$$(a) \quad 0 = h = \frac{\partial G}{\partial M} = 2a(T-T^*)M - cM^2 + dM^3 = 0 \quad \text{or } M=0$$

$$2a(T-T^*) - cM + dM^2 = 0$$

$$M = \frac{c \pm \sqrt{c^2 - 4d(T-T^*) \cdot 2a}}{2d} = \frac{c \pm \sqrt{c^2 - 8ad(T-T^*)}}{2d}$$

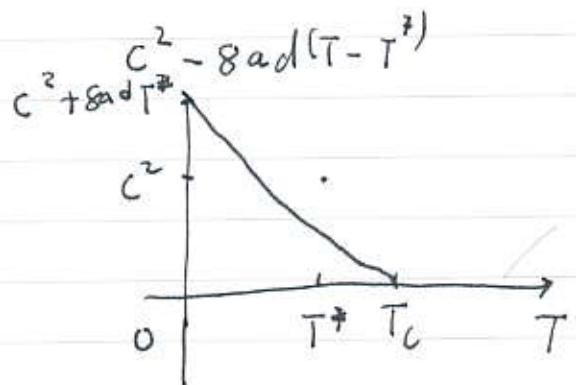
No real solution if $c^2 - 8ad(T-T^*) < 0$

$$\text{or } c^2 < 8ad(T-T^*)$$

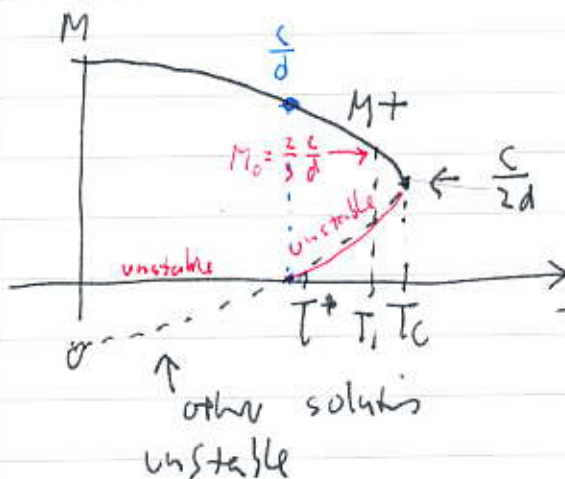
$$\text{or } \frac{c^2}{8ad} < T - T^*$$

$$\text{or } T_g = \frac{c^2}{8ad} + T^* \leq T$$

So $T - T^* \geq \frac{c^2}{8ad}$



There are two solutions for M



$$2a(T-T^*) = dM^2 - cM$$

$$= d\left(M - \frac{c}{2d}\right)^2 - \frac{c^2}{4d^2}$$



need to be more careful

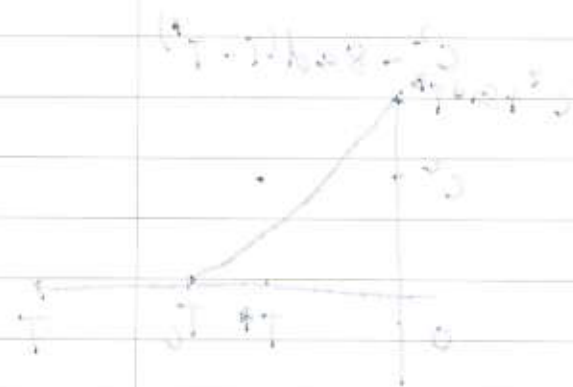
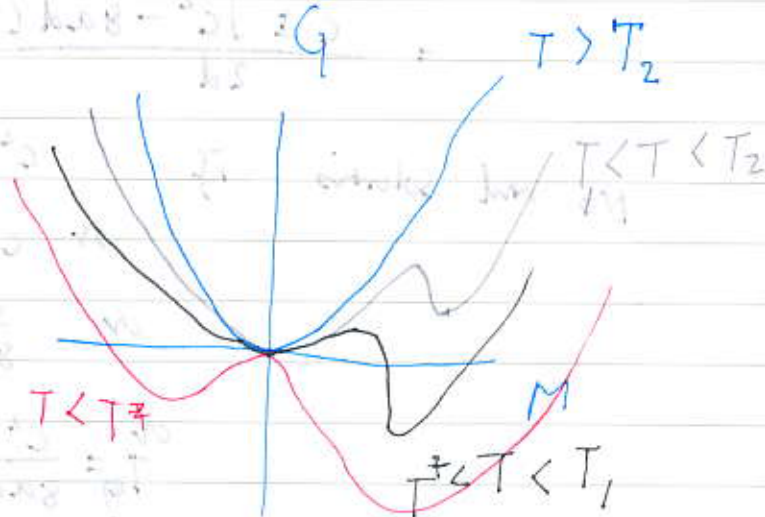
$M > 0$ has smaller G thus more stable

$$T_1 = T^* + \frac{1}{9} \frac{c^2}{d \cdot a}$$

$$Q = T_1 \Delta S = T_1 (10 \rightarrow a M_0^2)$$

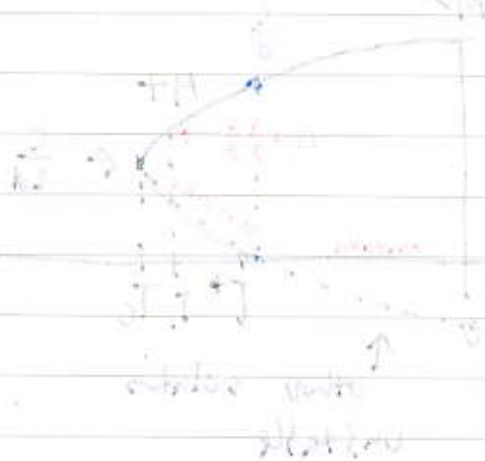
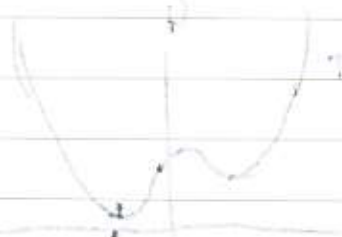
$$= \left(T^* + \frac{1}{9} \frac{c^2}{d \cdot a} \right) \left(a \cdot \left(\frac{2+c}{3d} \right)^2 \right)$$

$$= T_1 a \frac{c^2}{d^2} \left(\frac{4}{9} \right)$$



$$H_0 = (T^* - T) \Delta S$$

$$H_1 = (T^* - T) \Delta S$$



Handwritten notes at the bottom left, possibly describing the conditions or parameters of the graph.

Handwritten notes at the bottom right, possibly describing the conditions or parameters of the graph.

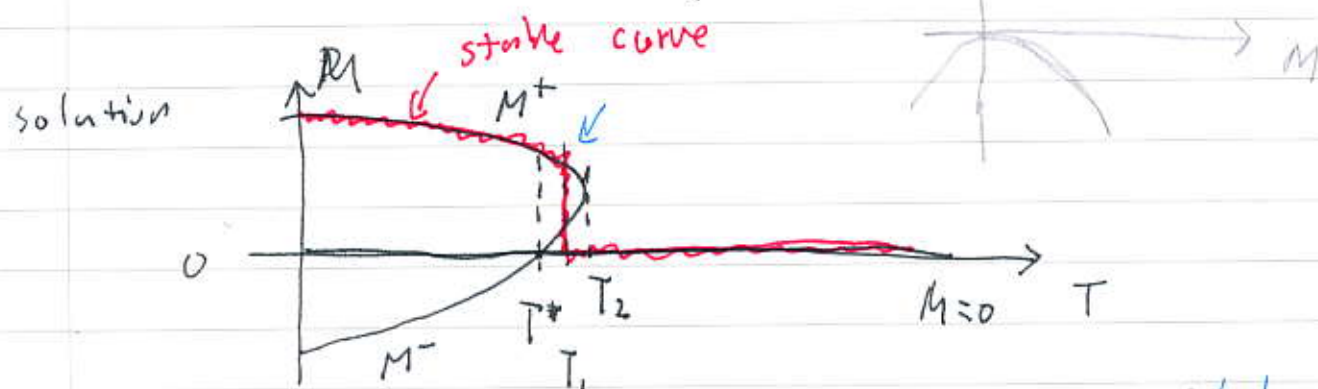
latent heat

$$Q = T_c \Delta S = T_c \Delta S \quad dQ = T dS$$

$$S = -\frac{\partial G}{\partial T} = -aM^2$$

$$\Delta S = S(T_c^+) - S(T_c^-) = 0 + aM^2 = a\left(\frac{c}{2d}\right)^2$$

$$Q = \left(\frac{c^2}{8ad} + T^*\right) a\left(\frac{c}{2d}\right)^2$$



We have 3 solutions need to decide which one has global min.

$$M=0 \text{ solution} \quad G = G_0 \neq \text{indep of } T$$

$$\text{for } M \neq 0 \quad G(M) = G(M=0)$$

$$G_0 + a(T-T^*)M_0^2 - \frac{1}{3}cM_0^3 + \frac{1}{4}dM_0^4 = G_0$$

value of M at the transition T_1

$$\begin{cases} a(T-T^*) - \frac{1}{3}cM_0 + \frac{1}{4}dM_0^2 = 0 \\ 2a(T-T^*) - cM_0 + dM_0^2 = 0 \end{cases}$$

but M_0 is also on the M curve

eliminate $T-T^*$

$$\left(-\frac{2}{3} + 1\right)cM_0 + \left(\frac{1}{2} - 1\right)dM_0^2 = 0$$

$$\frac{1}{3}cM_0 + \frac{1}{2}dM_0^2 = 0$$

$$M_0 = \frac{2c}{3d}$$

$$\frac{1}{9} \frac{c^2}{d}$$

$$a(T-T^*) = \frac{1}{3}c \frac{2c}{3d} - \frac{1}{4}d \left(\frac{2c}{3d}\right)^2 = \left(\frac{2}{9} - \frac{1}{4 \cdot 9}\right) =$$

$$M = \langle \sum_{i=1}^N \sigma_i \rangle =$$

low T expansion
ground state

$$\sum \sigma_i = N$$

$$\Delta E = E - E_0 = 0$$

ground

$$1$$

1st excited

$$\sum \sigma_i = N - 2$$

$$\Delta E = 8J$$

$$N$$

$$\sum \sigma_i = N - 4$$

$$\Delta E = 12J$$

$$2N$$

$$X = e^{-2K}$$

$$M = \frac{N \cdot 1 \cdot 1 + (N-2) e^{-8K} \cdot N + (N-4) e^{-12K} \cdot 2N}{1 + e^{-8K} N + \cancel{1N} e^{-12K} 2N}$$

$$= \frac{N + N(N-2) X^4 + (N-4) \cdot 2N X^6}{1 + N X^4 + 2N X^6 + \dots}$$

$$\frac{1}{1+X} = 1 - X + X^2 + \dots$$

$$= \cancel{N} + N$$

$$\frac{M}{N} = \left[1 + (N-2) X^4 + 2(N-4) X^6 \right] \left[1 - N X^4 - 2N X^6 + (N X^4 + 2N X^6)^2 + \dots \right]$$

$$= 1 + (N-2) X^4 + 2(N-4) X^6 - N X^4 - 2N X^6 + O(X^8)$$

$$= 1 - 2X^4 - 8X^6 + \dots$$

$$e^{\sinh(2K)} = \frac{e^{2K} - e^{-2K}}{2}$$

$$= \left(\frac{1}{X} - X \right)^{\frac{1}{2}}$$

$$\frac{M}{N} = \left(1 - \frac{1}{\sinh^4(2K)} \right)^{\frac{1}{8}}$$

$$= \left(1 - \frac{1}{16 \left(\frac{1}{X} - X \right)^4} \right)^{\frac{1}{8}}$$

$$= \left(1 - \frac{16}{\left[\frac{1}{X} (1-X^2) \right]^4} \right)^{\frac{1}{8}} = \left(1 - \frac{16 X^4}{(1-X^2)^4} \right)^{\frac{1}{8}}$$

$$= \left(1 - 16 X^4 (1 + X^2 + X^4 + X^6 + \dots)^4 \right)^{\frac{1}{8}}$$

$$\frac{M}{N} = (1 - 16x^4(1 + 4x^2 + \dots))^{\frac{1}{8}}$$

$$= (1 - 16x^4 - 16 \cdot 4x^6 + \dots)^{\frac{1}{8}}$$

$$\chi = e^{-\gamma K}$$

Date

$$= 1 - \frac{1}{8} \cdot 16x^4 - \frac{16 \cdot 4x^6}{8} + \dots = 1 - 2x^4 - 8x^6 + \dots$$

$$\quad \quad \quad -34x^8$$

$$2. (a) Z_{\sigma} = \sum_{\{\sigma\}} e^{-\beta \sum_{i=1}^N (1 - \varepsilon(-1)^i) \sigma_i \sigma_{i+1}}$$

$$-152x^{10}$$

$$\begin{array}{cccccc} & 1+\varepsilon & & 1-\varepsilon & & 1+\varepsilon & & 1-\varepsilon \\ | & | & | & | & | & | & | & | \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \end{array}$$

$$\underbrace{e^{\beta \sigma_1 \sigma_2 (1+\varepsilon)}}_P \underbrace{e^{\beta \sigma_2 \sigma_3 (1-\varepsilon)}}_Q e^{\beta \sigma_3 \sigma_4 (1+\varepsilon)} e^{\dots}$$

$$P = \begin{pmatrix} e^{\beta(1+\varepsilon)} & e^{-\beta(1+\varepsilon)} \\ e^{-\beta(1+\varepsilon)} & e^{\beta(1+\varepsilon)} \end{pmatrix} \quad Q = \begin{pmatrix} e^{\beta(1-\varepsilon)} & e^{-\beta(1-\varepsilon)} \\ e^{-\beta(1-\varepsilon)} & e^{\beta(1-\varepsilon)} \end{pmatrix}$$

(b)

$$\overline{F_{\sigma}} = -k_B T$$

$$Z_{\sigma} = \text{Tr}(PQ)^{\frac{N}{2}} = \frac{N}{2} \ln \lambda \quad \lambda^{\frac{N}{2}}$$

$$Z_{\text{tot}} = \int_{-\infty}^{+\infty} d\varepsilon e^{-\beta N \omega \varepsilon^2 + \frac{N}{2} \ln \lambda}$$

$$= \int_{-\infty}^{\infty} d\varepsilon e^{-\beta N \left(\omega \varepsilon^2 + \frac{-1}{2\beta} \ln \lambda \right)}$$

$$\quad \quad \quad \underbrace{\hspace{10em}}_{g(\varepsilon)}$$

$$g(\varepsilon) = \omega \varepsilon^2 - \frac{1}{2} k_B T \ln \lambda$$

$$PQ = \begin{pmatrix} x \cdot y & \frac{1}{x}y \\ \frac{1}{x}y & x \cdot y \end{pmatrix} \begin{pmatrix} x/y & y/x \\ y/x & x/y \end{pmatrix}$$

$$\text{let } x = e^{\beta} \\ y = e^{\beta \epsilon}$$

$$a = x^2 + \frac{1}{x^2} \\ b = y^2 + \frac{1}{y^2}$$

$$= \begin{pmatrix} x^2 + \frac{1}{x^2} & y^2 + \frac{1}{y^2} \\ \frac{1}{y^2} + y^2 & x^2 + \frac{1}{x^2} \end{pmatrix} = \begin{pmatrix} a & b \\ b & a \end{pmatrix}$$

$$\det \begin{pmatrix} a-\lambda & b \\ b & a-\lambda \end{pmatrix} = 0 \quad (a-\lambda)^2 - b^2 = 0$$

$$a-\lambda = \pm b$$

$$\lambda = a \pm b = (e^{2\beta} + e^{-2\beta}) \pm (e^{2\beta\epsilon} + e^{-2\beta\epsilon})$$

$$= 2(\cosh(2\beta) \pm \cosh(2\beta\epsilon)) \quad \cosh x \geq 0$$

+ is bigger than - 

$$\text{so } \lambda(\epsilon) = 2[\cosh(2\beta) + \cosh(2\beta\epsilon)]$$

$$\begin{bmatrix} a & \frac{1}{a} \\ \frac{1}{a} & a \end{bmatrix} \quad \begin{bmatrix} b & \frac{1}{b} \\ \frac{1}{b} & b \end{bmatrix}$$

for the row sum

$$g = \psi$$

$$(1 + \frac{1}{g}) \cdot \frac{1}{g} = 1 + \frac{1}{g^2} \cdot \frac{1}{g} = \frac{1}{g^2}$$

for the row sum

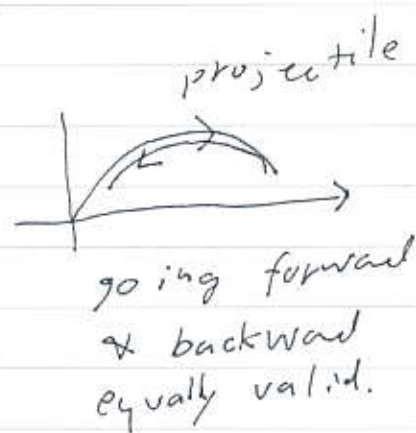
Microscopic laws are time-reversal ^{symmetric!} system

Newton's law $m_i \frac{d^2 \vec{r}_i}{dt^2} = \vec{F}(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N)$

if $\vec{r}_i = \vec{r}_i(t)$ is a solution

so is $\vec{r}_i(-t)$

Newton's Eq. is invariant under $t \rightarrow -t$

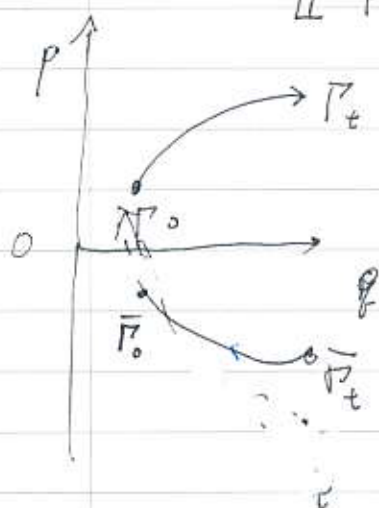


Hamiltonian Dynamics

$\Gamma_0 = (q_0, p_0)$ starts for Γ_0 we can move to $\Gamma_t = (q_t, p_t) = e^{tL} \Gamma_0$
after time t . $L = [\cdot, H]$ poisson bracket

Γ phase space

at Γ_t if we reverse the velocity. $p_t \rightarrow -p_t$



when $\bar{\Gamma}_0 = e^{tL} (q_t, -p_t) = (q_0, -p_0)$

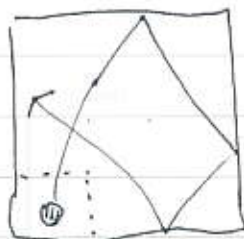
continues running in time forward

Quantum Mechanics

$\Psi(t)$ is a solution $\hat{H} \Psi = E \Psi$

then $\Psi^*(t)$ is also a solution a solution moves back in time

Same is true for electromagnetic Equations of Maxwell.

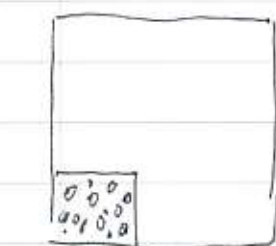


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one atom in a box at a box. $\rightarrow$

you cannot see a difference if the motion of the atom is reversed

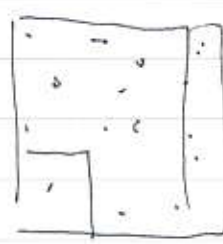
(a movie played forward or backward, there is no difference)



$\rightarrow V_0$ $t=0$
many atoms
of order 10^{23}



free expansion



$t \rightarrow \infty$

$$V > V_0$$

$H = \frac{U}{T} = \text{const}$
energy is conserved.
micro canonical

Boltzmann's principle

$$S = k \ln \Omega$$

Ω $\uparrow$ microscopic states

entropy (classical ideal gas)

$$S = Nk_B \ln V + \text{terms depend on } U$$

(const if U is fixed)

[recall Sackur-Tetrode formula]

entropy increases by $\Delta S = Nk_B \ln \frac{V}{V_0}$

Reverse situation never happens no matter how long you wait. why? (because $N \rightarrow \infty$)

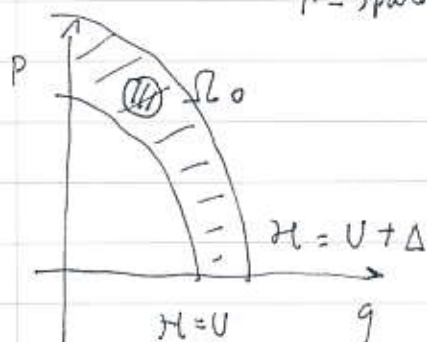
irreversibility due to the thermodynamic limit?

it is easy to see if there is one or two atoms, it is frequent that two atoms go back to small box V_0 often!

$$TdS \geq dU + PdV = \delta Q$$

$\uparrow$ equal sign holds only in reversible process

Poincaré Recurrence

 Γ -space Ω_n is the set

Theorem (1892)

(0) the phase flow is ~~chaotic~~ one-to-one & onto

(1) phase space volume is finite

(2) starting in small reg Ω_0 in phase space, because Liouville theorem the hypervolume does not change(3) $\Omega_n = \hat{T}_t \Omega_{n-1}$ for some t shown as later $\Omega_n \cap \Omega_0 \neq \emptyset$ (Consider $\Omega_0, \Omega_1, \dots, \Omega_n$)if they are all non intersecting, then as $n \rightarrow \infty$, the total phase volume goes to infinity

$$\sum_{i=0}^n |\Omega_i| = n |\Omega_0| \rightarrow \infty \text{ contradicts the}$$

assumption that the phase space is finite. (remember $S = k \ln \Omega$)

(in the sense of a set)

so the system returns to starting point arbitrarily close after t_p .However $t_p \propto V^N \rightarrow \infty$ as $N \rightarrow \infty$. t_p is too large (more than the age of Universe) ^{to be} relevant.

However, if entropy is the function of Γ , we have to conclude then S can ^{decrease} ~~increase~~ as well as increase due to Poincaré Recurrence!

Gibbs entropy

$$\bar{S} = -k_B \int p(\Gamma) \ln p(\Gamma) d\Gamma$$

$$\int p(\Gamma) d\Gamma = 1$$

Let assume we have a microscopic ensemble

$$p(\Gamma) = \begin{cases} C & U < H < U + \Delta \\ 0 & 0 \end{cases}$$

$$C \int d\Gamma = C \Omega = 1$$

$$U < H < U + \Delta$$

$$\text{then } \bar{S} = -k_B \int C \ln C d\Gamma = -k_B \ln C = k_B \ln \Omega$$

$$U < H < U + \Delta$$

consistent with Boltzmann entropy.

in equilibrium.

 $\bar{S}$ do not increase even in nonequilibrium situation $\bar{S} = \text{const}$ for any $p(\Gamma, t)$ function + p are the
 $p(\Gamma, t) = p(\Gamma)$

prove

$$\frac{d\bar{S}}{dt} = -k_B \int \frac{\partial}{\partial t} (p \ln p) d\Gamma$$

$$= -k_B \int \left[\frac{\partial p}{\partial t} \ln p + p \frac{\partial \ln p}{\partial t} \right] d\Gamma$$

$$\frac{\partial \ln p}{\partial t} = \frac{1}{p} \frac{\partial p}{\partial t}$$

$$= -k_B \int \left(\frac{\partial p}{\partial t} \right) (\ln p + 1) d\Gamma$$

$$\text{but } \frac{\partial p}{\partial t} + \{p, H\} = 0$$

Poisson bracket

$$\frac{\partial p}{\partial t} + \frac{\partial p}{\partial q} \frac{\partial H}{\partial p} - \frac{\partial p}{\partial p} \frac{\partial H}{\partial q} = 0$$

$$= -k_B \int \left[-\frac{\partial p}{\partial q} \dot{q} + \frac{\partial p}{\partial p} \dot{p} \right] (\ln p + 1) d\Gamma$$

$$\frac{d\bar{S}}{dt} = -k_B \int \frac{\partial}{\partial t} (p \ln p) d\Gamma = -k_B \int \frac{D}{Dt} (p \ln p) d\Gamma = 0$$

because $\frac{Dp}{Dt} = 0$

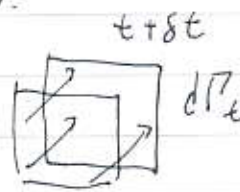
by Liouville theorem

$$\int \frac{\partial}{\partial t} g d\Gamma \equiv \int \frac{D}{Dt} g d\Gamma$$

show this more carefully.



fixed
volume element
now # of
system change
over time



g does not change
follow the flow
interior all in
so $\int g d\Gamma$ will not change

how g change
following the motion
of particle.

however $d\Gamma_t = d\Gamma$
by Liouville theorem

$\bar{S}$ should not change for our gas experiment!

Same is true if we use QM in $\hat{\rho}$ density matrix

$$\bar{S} = \text{Tr}(\hat{\rho} \ln \hat{\rho})$$

$$\frac{d\bar{S}}{dt} \equiv 0$$

This appears to say entropy
is only meaningful for Equilibrium macroscopic system!!

proof of

H-theorem

one particle distribution valid only
for dilute gas

$$\rightarrow S_B = -k_B \int f \ln f d\vec{r} d\vec{p}$$

$$= -k_B H$$

Boltzmann's version
for reduced number f

$$f(\vec{r}_1) \approx N \int \rho(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N, \vec{p}_1, \dots, \vec{p}_N) d\vec{r}_2 \dots d\vec{r}_N d\vec{p}_2 \dots d\vec{p}_N$$

$$H = \int f \ln f \, d\vec{r} d\vec{p}$$

$$\frac{dH}{dt} = \int \frac{\partial}{\partial t} [f \ln f] \, d\vec{r} d\vec{p} = \int \left(\frac{\partial f}{\partial t} \ln f + \frac{\partial f}{\partial t} \right) d\vec{r} d\vec{p}$$

Date

NO contrib to int

$$= \int \frac{\partial f}{\partial t} (1 + \ln f) \, d\vec{r} d\vec{p}$$

$\vec{v}$ and $\vec{r}$ are independent $\vec{v} = \frac{\vec{p}}{m}$

$$\frac{\partial f}{\partial t} + \vec{v} \cdot \frac{\partial f}{\partial \vec{r}} + \vec{F} \cdot \frac{\partial f}{\partial \vec{p}} = \left(\frac{\partial f}{\partial t} \right)_{\text{coll}}$$

consider

$$\int \frac{\partial f}{\partial x} (v_x f \ln f) \, d\vec{r} d\vec{p}$$

$$= \int \frac{\partial}{\partial x} (v_x f \ln f) \, d\vec{r} d\vec{p} = \int dx \frac{\partial}{\partial x} v_x f \ln f \, dy dz d\vec{p}$$

$$= \left[v_x f \ln f \right]_{x=-\infty}^{x=\infty} = 0$$

density $f=0$ at the wall

$$\frac{\partial}{\partial x} (f \ln f) = \frac{\partial f}{\partial x} \ln f + f \frac{1}{f} \frac{\partial f}{\partial x}$$

$$= \frac{\partial f}{\partial x} (1 + \ln f)$$

same trick works for $\frac{\partial F_x f}{\partial p_x}$

assume $f \rightarrow 0$ when $|x| \rightarrow \infty$ or $|p| \rightarrow \infty$ as long as F is a function of $\vec{r}$ only (i.e. conservative force)

→ so

$$\frac{dH}{dt} = \int (1 + \ln f) \left(\frac{\partial f}{\partial t} \right)_{\text{coll}} d\vec{r} d\vec{p}$$

 $\vec{p} = p$ to save writing

$$\left(\frac{\partial f}{\partial t} \right)_{\text{coll}} = \int d\vec{p}_1 \int d\vec{p}'_1 d\vec{p}'_2 \sigma(\vec{p}, \vec{p}_1 | \vec{p}'_1 \vec{p}'_2) (f'_1 f'_2 - f f_1)$$

$$\frac{dH}{dt} = \int d\vec{r} d\vec{p} d\vec{p}_1 \int d\vec{p}'_1 d\vec{p}'_2 \sigma(\vec{p}, \vec{p}_1 | \vec{p}'_1 \vec{p}'_2) \cancel{(1 + \ln f)} \cancel{(f'_1 f'_2 - f f_1)}$$

$\cancel{1 + \ln f}$

$$\frac{dH}{dt} = \int d\vec{r} \int dP \int dP_1 \int dP' \int dP'_1 \underbrace{\sigma(P P_1 | P' P'_1)}_{\substack{\text{symmetric due to} \\ \text{permutation of} \\ \text{prime}}} (1 + \ln f) (f' f'_1 - f f_1)$$

all P 's are dummy integration variables

$$\int dP dP_1 dP' dP'_1 \sigma(P P_1 | P' P'_1) f(P') f(P'_1) \leftarrow \text{interchange}$$

$$\int dP' dP'_1 dP dP_1 \sigma(P' P'_1 | P P_1) f f_1$$

$P \rightarrow P'$
 $P_1 \rightarrow P'_1$

this becomes identical to the second "term".
need time-reversed plus space inversion

So

$$\frac{dH}{dt} = \int d\vec{r} \int dP dP_1 dP' dP'_1 \sigma(P P_1 | P' P'_1) \ln f (f' f'_1 - f f_1)$$

interchange $P \leftrightarrow P'$, $P_1 \leftrightarrow P'_1$

$$\textcircled{1} \sigma(P P_1 | P' P'_1) \ln f (f' f'_1 - f f_1) = \Phi_1 = \Phi$$

$$\textcircled{2} \sigma(P_1 P | P'_1 P') \ln f_1 (f'_1 f' - f_1 f) = \Phi_2 = \Phi$$

interchange prime & non-prime

$$\textcircled{3} \sigma(P' P'_1 | P P_1) \ln f' (f f_1 - f' f'_1) = \Phi_3 = \Phi$$

$$\textcircled{4} \sigma(P'_1 P'_1 | P_1 P) \ln f'_1 (f_1 f - f'_1 f') = \Phi_4 = \Phi$$

above

$\{ P \leftrightarrow P', P_1 \leftrightarrow P'_1 \}$ we all the same by symmetry $\ln 1 = 0$

add all $\Phi = \frac{1}{4} (\Phi_1 + \Phi_2 + \Phi_3 + \Phi_4)$

$$\frac{dH}{dt} = \frac{1}{4} \int d\vec{r} \int dP dP_1 dP' dP'_1 \sigma(P P_1 | P' P'_1) \left[\ln \left(\frac{f f_1}{f' f'_1} \right) \cdot (f' f'_1 - f f_1) \right]$$

$$= \int d\vec{r} \int dP dP_1 dP' dP'_1 \sigma(\dots) (B-A) \ln \frac{A}{B}$$

$\sigma \geq 0, A \geq 0, B \geq 0$

if $A < B$, $B-A > 0$, $\ln \frac{A}{B} < 0$ $(B-A) \ln \frac{A}{B} < 0$
 $A > B$ $B-A < 0$, $\ln \frac{A}{B} > 0$ $(B-A) \ln \frac{A}{B} < 0$

so $\frac{dH}{dt} \leq 0$ $S_B \geq 0$

H-theorem $\frac{dH}{dt} \leq 0$

i.e. $\int f \ln f d\vec{r} d\vec{p}$ is constant or decrease but never increase with time.

Entropy then always non-decreasing. Entropy increases.

"Entropy cannot be a microscopic quantity"

Entropy concept appears (in the sense $S > 0$) only at coarse-grain level. (macroscopic level)

Boltzmann Eq. describes irreversible process

For Maxwell distribution

$$f = A e^{-\beta \frac{p^2}{2m}}$$

- 1) from thermodynamic point of view entropy is unique & physical (because it is heat)

$$\Delta S = \frac{\delta Q}{T} \text{ in revers. process}$$

- 2) Then is stat-mech entropy depends on the level of coarsening?

i.e. Maxwell distribution is stationary.

equilibrium state

$$\frac{dH}{dt} = 0$$

$$f_0 f_1 - f'_0 f'_1 = 0$$

is a solution

if f is Maxwell distribution then $\frac{dH}{dt} = 0$

- 3) what level of coarsening is good enough?

Ans:

due to energy conservation

$$\frac{f f_1}{f' f'_1} = e^{-\beta \left(\frac{p^2}{2m} + \frac{p_1^2}{2m} - \frac{p'^2}{2m} - \frac{p'_1^2}{2m} \right)} = 1$$

$$\text{so } \frac{dH}{dt} = 0$$

i.e. coarse-grain Gibbs formula

Maxwell distribution

Renormalization group transformation in 1D

Date

$|a|$
 0 x 0 x 0 x 0 x 0
 1 2 3 4 5 6 7 8 9 ... before transfer

$\overbrace{1 \dots 3}^{2a}$
 $2a$

 $b=2$

decimation

 N spins

0 0 0 0 0
 1 3 5 7 ...
 ↓ ↙ ↘ ↙ ↘
 0 0 0 0 ...

scale the system
back to lattice
constant a

renam.

1 2 3 ... $\frac{N}{2}$ spins

Real space RG

- ① find out the transform of H
- ② find fixed pt and line and fixed pt to find critical exponents

$$\bar{H} = -K \sum_{i=1}^N \sigma_i \sigma_{i+1} - h \sum_{i=1}^N \sigma_i - \sum_{i=1}^N C \leftarrow \text{free energy}$$

β absorbed in couplings

$$Z = \sum_{\{\sigma\}} e^{-\bar{H}}$$

Aim: to change the scale by a factor of 2.

trace out the even-numbered spins.

$$Z = \sum_{\{\sigma\}} e^{\left(K \sum_{i=1}^N \sigma_i \sigma_{i+1} + h \sum_{i=1}^N \sigma_i + \sum_{i=1}^N C \right)}$$

$$= \sum_{\{\sigma\}} e^{K \sigma_1 \sigma_2 + h \sigma_1 + C} e^{K \sigma_2 \sigma_3 + h \sigma_2 + C} e^{K \sigma_3 \sigma_4 + h \sigma_3 + C} \dots$$

$$Z = \sum_{\{\sigma\}} \prod_{i=2,4,6} e^{\{K \sigma_i (\sigma_{i-1} + \sigma_{i+1}) + h \sigma_i + \frac{1}{2} h (\sigma_{i-1} + \sigma_{i+1}) + 2c\}}$$

sum over σ_i for i even



$$= \sum_{\sigma_1 \sigma_3 \dots \sigma_{i-1}} \prod_{i=2,4,6} \left(e^{K(\sigma_{i-1} + \sigma_{i+1}) + h + \frac{h}{2} (\sigma_{i-1} + \sigma_{i+1}) + 2c} + e^{-K(\sigma_{i-1} + \sigma_{i+1}) - h + \frac{h}{2} (\sigma_{i-1} + \sigma_{i+1}) + 2c} \right)$$

Relabeling the spin

$$Z' = \sum_{\{\sigma\}} \prod_{i=1}^{N/2} \left(e^{(K + \frac{h}{2})(\sigma_i + \sigma_{i+1}) + h + 2c} + e^{(-K + \frac{h}{2})(\sigma_i + \sigma_{i+1}) - h + 2c} \right)$$

We demand the renormalized partition function can be cast in the same form as the original

$$= \sum_{\{\sigma\}} \prod_i e^{(K' \sigma_i \sigma_{i+1} + h' \sigma_i + c')} \quad i=1$$

symmetric form

$$e^{K' \sigma_1 \sigma_2 + \frac{h'}{2} (\sigma_1 + \sigma_2) + c'}$$

$$= e^{(K + \frac{h}{2})(\sigma_1 + \sigma_2) + h + 2c} + e^{(-K + \frac{h}{2})(\sigma_1 + \sigma_2) - h + 2c}$$

$$\left. \begin{array}{l} \sigma_1 = \sigma_2 = +1 \\ \sigma_1 = \sigma_2 = -1 \\ \sigma_1 = -\sigma_2 = +1 \\ \sigma_1 = -\sigma_2 = -1 \end{array} \right\} \begin{cases} e^{K' + h' + c'} = e^{(2K + h + 2c)} + e^{(-2K + 2c)} & (1) \\ e^{K' - h' + c'} = e^{(2K - h + 2c)} + e^{(-2K + 2c)} & (2) \\ e^{-K' + c'} = e^{h + 2c} + e^{-h + 2c} & (3) \end{cases}$$

$$(1) \div (2) \quad e^{2h'} = \frac{e^{2h} e^{2K} + e^{-2K}}{e^{-2h} e^{2K} + e^{-2K}} = e^{2h} \frac{\cosh(2K + h)}{\cosh(2K - h)}$$

This means

$$\bar{f}_s(x, \epsilon) = b^{-1} \tilde{f}_s(b^2 x, b\epsilon) \quad \leftarrow d=1$$

The usual variable has to be replaced by $x = e^{-4K}$!

Date

Q5.
$$\begin{cases} m \frac{dv}{dt} = - \frac{dV}{dx} - m \int_{-\infty}^t \gamma(t-t') v(t') dt' + R(t) \\ \frac{dx}{dt} = v \end{cases} \quad x, v$$

$$(a) \quad \frac{d}{dt} E(t) = \frac{d}{dt} \left(\frac{1}{2} m v^2 + V \right) = \frac{1}{2} m 2v \frac{dv}{dt} + \frac{dV}{dx} \frac{dx}{dt}$$

$$= m v \dot{v} + \frac{dV}{dx} v$$

$$= v \left[- \frac{dV}{dx} - m \int_{-\infty}^t \gamma(t-t') v(t') dt' + R(t) \right] + \frac{dV}{dx} v$$

$$= v(t) \left[- m \int_{-\infty}^t \gamma(t-t') v(t') dt' + R(t) \right]$$

$$(b) \quad W_p = \int_0^T m v(t) \int_{-\infty}^t \gamma(t-t') v(t') dt' dt$$

$$W_R = \int_0^T v(t) R(t) dt$$

$$\int_0^T m \frac{dv}{dt} \cdot v dt = - \int_0^T \frac{dV}{dx} \cdot v(t) dt \oplus W_p + W_R$$

$$\int_0^T \left[m v \dot{v} + \frac{dV}{dx} \cdot v \right] dt = -W_p + W_R$$

$$= \int_0^T \frac{dE(t)}{dt} dt = \int_0^T dE(t) = E(T) - E(0)$$

$\leftarrow$ some
bound
value

$$W_R \propto T, \quad W_p \propto T,$$

More precisely $\frac{1}{T} [E(T) - E(0)] = -\frac{W_0}{T} + \frac{W_R}{T} \rightarrow 0$ as $T \rightarrow \infty$ 291

power = $\frac{\text{work}}{\text{time}}$

$\langle v \cdot R \rangle = \langle v_m \rangle \int_{-\infty}^t \gamma(t-t') v(t') dt'$

$\langle \rangle = \frac{1}{T} \int_0^T dt$

average rate of power-dissipation

problem 6.3)

consider $\frac{d}{d\lambda} (e^{\lambda H} A e^{-\lambda H}) = e^{\lambda H} H A e^{-\lambda H} + e^{\lambda H} (A(-H)) e^{-\lambda H}$
 $= e^{\lambda H} (HA - AH) e^{-\lambda H} = e^{\lambda H} [H, A] e^{-\lambda H} = e^{\lambda H} (-i\hbar \dot{A}) e^{-\lambda H}$
 $= -i\hbar e^{\lambda H} \dot{A} e^{-\lambda H}$

$i\hbar \frac{dA}{dt} = [A, H]$
 $= i\hbar \dot{A}$
 Heisenberg eq

integrate both sides $\int_0^\beta d\lambda$

$\int_0^\beta d\lambda \frac{d}{d\lambda} (e^{\lambda H} A e^{-\lambda H}) = -i\hbar \int_0^\beta d\lambda e^{\lambda H} \dot{A} e^{-\lambda H}$

$e^{\beta H} A e^{-\beta H} - A = -i\hbar \int_0^\beta d\lambda e^{\lambda H} \dot{A} e^{-\lambda H}$

multiply ρ for left
 B for right
 take trace

$\beta \langle \dot{A}; B \rangle = \frac{1}{\beta} \int_0^\beta d\lambda \text{Tr}(\rho e^{\lambda H} \dot{A} e^{-\lambda H} B)$

$\rho = \frac{e^{-\beta H}}{Z} = \left(\frac{-1}{i\hbar} \right) \text{Tr}[\rho e^{\beta H} A e^{-\beta H} - \rho A B]$

$= \frac{-1}{i\hbar} \text{Tr} \left[\frac{1}{Z} A e^{-\beta H} B - \rho A B \right]$

$= \frac{-1}{i\hbar} \text{Tr}[\rho B A - \rho A B] = \frac{1}{i\hbar} \text{Tr}[\rho (A B - B A)]$
 $= \frac{1}{i\hbar} \langle [A, B] \rangle$

left $\begin{matrix} \rightarrow & \rightarrow & \leftarrow & \leftarrow \\ \rightarrow & \leftarrow & \leftarrow & \leftarrow \end{matrix}$ right

$$P = \begin{pmatrix} 1 & e^{-\beta \epsilon} & e^{-\beta \epsilon} & 0 \\ e^{-\beta \epsilon} & 1 & 1 & e^{-\beta \epsilon} \\ e^{-\beta \epsilon} & 1 & 1 & e^{-\beta \epsilon} \\ 0 & e^{-\beta \epsilon} & e^{-\beta \epsilon} & 1 \end{pmatrix}$$

Date

or

$$P = \begin{pmatrix} 1 & x & x & 0 \\ x & 1 & 1 & x \\ x & 1 & 1 & x \\ 0 & x & x & 1 \end{pmatrix} \quad x = e^{-\beta \epsilon}$$

$$|P - \lambda I| = 0$$

$$\lambda = 1, 0, \frac{1}{2}(3 \pm \sqrt{1+16x^2})$$

$$\lambda_{\max} = \frac{1}{2}(3 + \sqrt{1+16x^2})$$

$$\begin{aligned} |P - \lambda I| &= \lambda^4 - 4\lambda^3 - 4x^2\lambda^2 + 5\lambda^2 + 4x^2\lambda - 2\lambda \\ &= (\lambda-1)\lambda \cdot (2 - 4x^2 - 3\lambda + \lambda^2) \end{aligned}$$

$$F = -k_B T \ln Z$$

$$= -k_B T N \ln \lambda_{\max}$$

prob 2.

(a)

$$\xi(t, 0) = b \xi(b^Y t, 0)$$

$$\text{Let } b^Y t = 1$$

$$b = t^{-\frac{1}{Y}}$$

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$$= t^{-\frac{1}{Y}} \dots$$

$$\text{compare with } \xi \propto t^{-\nu} \rightarrow \nu = \frac{1}{Y}$$

(b)

$$M = \langle \sum_i \sigma_i \rangle = \frac{1}{Z} \sum_{\{\sigma\}} \sum_i \sigma_i e^{-\beta(E(\sigma) - h \sum_i \sigma_i)}$$

$$\chi = \left. \frac{\partial M}{\partial h} \right|_{h \rightarrow 0} = \frac{1}{Z} \sum_{\{\sigma\}} \left(\sum_i \sigma_i \right) \left(\beta \sum_j \sigma_j \right) e^{-\beta H(\sigma)} - \frac{\langle \sum_i \sigma_i \rangle \langle \beta \sum_j \sigma_j \rangle}{Z^2}$$

$$= \beta \sum_{i,j} \langle \sigma_i \sigma_j \rangle - \langle \sigma_i \rangle \langle \sigma_j \rangle$$

$$= \beta \sum_{i,j} G(|i-j|) = \beta \sum_i \sum_j G(j-i)$$

$$= \beta N \int d\vec{r} G(\vec{r})$$

(c)

$$\chi = \beta N \int d\vec{r} G(r)$$

b

$$\text{let } b = \xi$$

$$= \beta N \int r^{d-1} dr b^{(-d+2-n)} G(r/b, \xi/b)$$

$$= \beta N \xi^{-d+2-n} \int r^{d-1} dr G\left(\frac{r}{\xi}, 1\right)$$

$$= \beta N \xi^{-d+2-n} \xi^d \int_0^\infty \left(\frac{r}{\xi}\right)^{d-1} d\left(\frac{r}{\xi}\right) G\left(\frac{r}{\xi}, 1\right)$$

$$\approx \xi^{2-n} = t^{-\nu(2-n)}$$

$$\text{compare with definition } \chi \approx t^{-\gamma}$$

$$\text{we find } \gamma = \nu(2-n) !$$

$$3. \quad m \frac{dv}{dt} = -m\gamma v + f + R(t)$$

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$$(a) \quad m \frac{d\langle v \rangle}{dt} = -m\gamma \langle v \rangle + f$$

$$\frac{d\langle v \rangle}{dt} = -\gamma \langle v \rangle + f/m$$

$$\frac{d\langle v \rangle}{-\gamma \langle v \rangle + f/m} = dt$$

$$-\frac{1}{\gamma} d(\ln(-\gamma \langle v \rangle + \frac{f}{m})) = dt$$

$$\ln(-\gamma \langle v \rangle + \frac{f}{m}) = -\gamma t + c$$

$$-\gamma \langle v \rangle + \frac{f}{m} = A e^{-\gamma t}$$

$$\langle v \rangle = \frac{f}{\gamma m} + A' e^{-\gamma t}$$

$$\langle v \rangle_0 = \frac{f}{\gamma m} + A'$$

$$\langle v \rangle = \frac{f}{\gamma m} + [\langle v \rangle_0 - \frac{f}{\gamma m}] e^{-\gamma t}$$

if $R(t) \neq 0$

$$\langle v \rangle(t) = \frac{f}{\gamma m} + [\langle v \rangle_0 - \frac{f}{\gamma m}] e^{-\gamma t} + e^{-\gamma t} \int_0^t \frac{R(\tau)}{m} e^{\gamma \tau} d\tau$$

$$m \frac{d\langle v \rangle(t)}{dt} = f e^{-\gamma t} - \gamma e^{-\gamma t} \int_0^t \frac{R(\tau)}{m} e^{\gamma \tau} d\tau + R(t)$$

$$= -m\gamma (\langle v \rangle - \frac{f}{m\gamma}) + R(t)$$

$$v(t) - \bar{v}(t) = e^{-\gamma t} \int_0^t \frac{R(\tau)}{m} e^{\gamma \tau} d\tau$$

$$\begin{aligned} \langle (v(t) - \bar{v}(t))^2 \rangle &= e^{-2\gamma t} \int_0^t d\tau d\tau' \frac{R(\tau) R(\tau')}{m^2} e^{\gamma(\tau + \tau')} \\ &= e^{-2\gamma t} \int_0^t d\tau d\tau' \frac{1}{m^2} C \delta(\tau - \tau') e^{\gamma(\tau + \tau')} \\ &= \frac{C}{m^2} e^{-2\gamma t} \int_0^t d\tau e^{2\gamma \tau} \\ &= \frac{C}{m^2} e^{-2\gamma t} \left[\frac{1}{2\gamma} e^{2\gamma \tau} \right]_0^t \\ &= \frac{C}{2\gamma m^2} [e^{2\gamma t} - 1] \\ &= \frac{C}{2\gamma m^2} [1 - e^{-2\gamma t}] \end{aligned}$$

$$C = 2\gamma m k_B T$$

$$\left[\frac{k_B T}{m} \right]$$

(3) Fokker-Planck equation

$$\frac{\partial P}{\partial t} = -\frac{\partial}{\partial v} (M_1 P) + \frac{1}{2} \frac{\partial^2}{\partial v^2} (M_2 P)$$

$$\begin{aligned} M_1 &= \frac{1}{t} \langle v(\tau) - v(0) \rangle & \tau \rightarrow 0 &= \frac{1}{t} \left[\frac{f}{\gamma m} + (v(0) - \frac{f}{\gamma m})(1 - e^{-\gamma t}) - v(0) \right] \\ M_2 &= \frac{1}{t} \langle (v(\tau) - v(0))^2 \rangle & &= \frac{1}{t} \left[\frac{f^2}{\gamma^2 m^2} + \gamma - \frac{f}{\gamma m} (v - \frac{f}{\gamma m}) - (v - \frac{f}{\gamma m}) \gamma t \right] \\ &= \frac{1}{t} \frac{C}{2\gamma m^2} [2\gamma t + \dots] & &= - (v - \frac{f}{\gamma m}) \cdot \gamma \\ &= \frac{C}{m^2} & & \end{aligned}$$

$$\frac{\partial P}{\partial t} = \frac{\partial}{\partial v} \left[\left(v - \frac{f}{\gamma m} \right) P \right] + \frac{1}{2} \frac{C}{m^2} \frac{\partial^2 P}{\partial v^2}$$

$$\mathcal{Z} = \sum_{n_1, n_2, \dots, n_N} e^{-\beta(\sum \epsilon n_i - \mu n_i)}$$

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$$= \sum_{n_1} e^{-\beta(\epsilon - \mu)n_1} \sum_{n_2} e^{-\beta(\epsilon - \mu)n_2} \dots$$

$$= \left[\sum_{n=0}^{\infty} e^{-\beta(\epsilon - \mu)n} \right]^N = \left[\frac{1}{1 - e^{-\beta(\epsilon - \mu)}} \right]^N$$

$$S = \frac{\partial}{\partial T} (k_B T \ln \mathcal{Z}) = \frac{\partial}{\partial T} (k_B T N \ln(1 - e^{-\beta(\epsilon - \mu)}))$$

$$\cancel{-k_B N} = -N k_B \ln(1 - e^{-\beta(\epsilon - \mu)})$$

$$- k_B T N \frac{e^{-\beta(\epsilon - \mu)}}{1 - e^{-\beta(\epsilon - \mu)}} \left[\frac{\epsilon - \mu}{k_B T^2} \right]$$

$$= -N k_B \ln(1 - e^{-\beta(\epsilon - \mu)}) - N \frac{(\epsilon - \mu)}{T} \frac{1}{e^{\beta(\epsilon - \mu)} - 1}$$

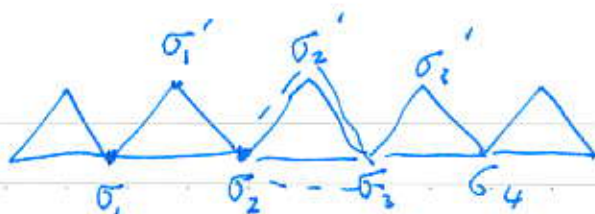
$$\langle N \rangle = \frac{\partial}{\partial \mu} (k_B T \ln \mathcal{Z}) = \cancel{k_B T}$$

$$= \frac{\partial}{\partial \mu} [N k_B T \ln(1 - e^{-\beta(\epsilon - \mu)})]$$

$$= \cancel{-N k_B T} \frac{-e^{-\beta(\epsilon - \mu)}}{1 - e^{-\beta(\epsilon - \mu)}} \cdot \cancel{\beta}$$

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3.



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$$Z = \sum_{\{\sigma\}} e^{K(\sigma_2\sigma_3 + \sigma_2\sigma_2' + \sigma_3\sigma_2')} e^{K(\sigma_3\sigma_4 + \sigma_3\sigma_3' + \sigma_4\sigma_4')}$$

$$= \sum_{\sigma_2'} \underbrace{\sum_{\sigma_2} e^{K(\sigma_2\sigma_3 + \sigma_2\sigma_2' + \sigma_3\sigma_2')}}_{P(\sigma_2\sigma_3)}$$

$$=$$



$$P(\sigma_2\sigma_3) = \sum_{\sigma_2'} e^{K(\sigma_2\sigma_3 + \sigma_2\sigma_2' + \sigma_3\sigma_2')}$$

$$= e^{K(\sigma_2\sigma_3 + \sigma_2 + \sigma_3)} + e^{K(\sigma_2\sigma_3 - \sigma_2 - \sigma_3)}$$

$$P = \begin{pmatrix} e^{3K} + e^{-K} & 2e^{-K} \\ 2e^{-K} & e^{3K} + e^{-K} \end{pmatrix}$$

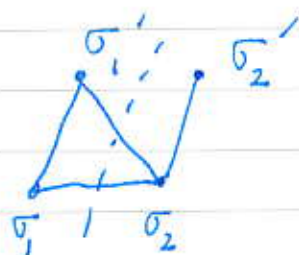
$$\begin{vmatrix} e^{3K} + e^{-K} - \lambda & 2e^{-K} \\ 2e^{-K} & e^{3K} + e^{-K} - \lambda \end{vmatrix} = 0$$

$$(e^{3K} + e^{-K} - \lambda)^2 - 4e^{-2K} = 0$$

$$\lambda = e^{3K} + e^{-K} \pm 2e^{-K} = \begin{cases} e^{3K} + 3e^{-K} \\ e^{3K} - e^{-K} \end{cases}$$

$$F = -Nk_B T \ln Z = -Nk_B T \ln \lambda$$

$$\lambda = e^{3K} + 3e^{-K}$$



$$P(\sigma_1, \sigma_1'; \sigma_2, \sigma_2') = e^{K(\frac{1}{2}\sigma_1\sigma_1' + \frac{1}{2}\sigma_2\sigma_2' + \sigma_1\sigma_2 + \sigma_1'\sigma_2')}$$

$$= \begin{matrix} \sigma_1\sigma_1' \\ \begin{matrix} + + \\ + - \\ - + \\ - - \end{matrix} \end{matrix} \begin{matrix} \sigma_2\sigma_2' \\ \begin{matrix} + + & + - & - + & - - \end{matrix} \end{matrix} \begin{bmatrix} e^{3K} & e^{2K} & e^{-2K} & e^{-K} \\ 1 & e^{-K} & e^K & 1 \\ 1 & e^K & e^{-K} & 1 \\ e^{-K} & e^{-2K} & e^{2K} & e^{3K} \end{bmatrix}$$

$$\lambda_1 = 0, \lambda_2 = 0, \lambda_3 = e^{3K} - e^{-K}, \lambda_4 = e^{3K} + 3e^{-K}$$

same as above

$$G = (T - T_c) a M^2 + b M^4$$

$$h = \frac{\partial G}{\partial M} = (T - T_c) 2a M + 4b M^3$$

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$$\langle \frac{1}{2} m v^2 \rangle = \frac{k_B T}{2}$$

$$5. (a) \quad P = \langle m \gamma v \cdot v \rangle = \gamma \langle m v^2 \rangle = \gamma k_B T$$

$$(b) \quad I = \langle R(t) v(t) \rangle$$

$$v(t) = v(0) e^{-\gamma t} + \int_0^t \frac{R(\tau)}{m} e^{-\gamma(t-\tau)} d\tau$$

 $t \rightarrow \infty$

$$I = \int_0^t \underbrace{\langle R(t) R(\tau) \rangle}_{\delta(t-\tau) \text{ or } \delta(\tau-t)} \frac{e^{-\gamma(t-\tau)}}{m} d\tau$$



we have to think of δ is an idealized gaussian-like peak

integrate to t take half of the value

$$= \frac{1}{m} \int_0^t \delta(\tau-t) e^{-\gamma(t-\tau)} d\tau = \frac{1}{2m}$$

(c) $\int$ fluctuation-dissipation give $\gamma = 2 m \gamma k_B T$
This shown $P = I$.

alternatively

$$m \frac{dv}{dt} = -m \gamma v + R(t)$$

$$\frac{d}{dt} \left(\frac{1}{2} m v^2 \right) = \dots$$

$$= \frac{1}{T} \int_0^T (-m \gamma v^2) dt + \frac{1}{T} \int_0^T R(t) v dt$$

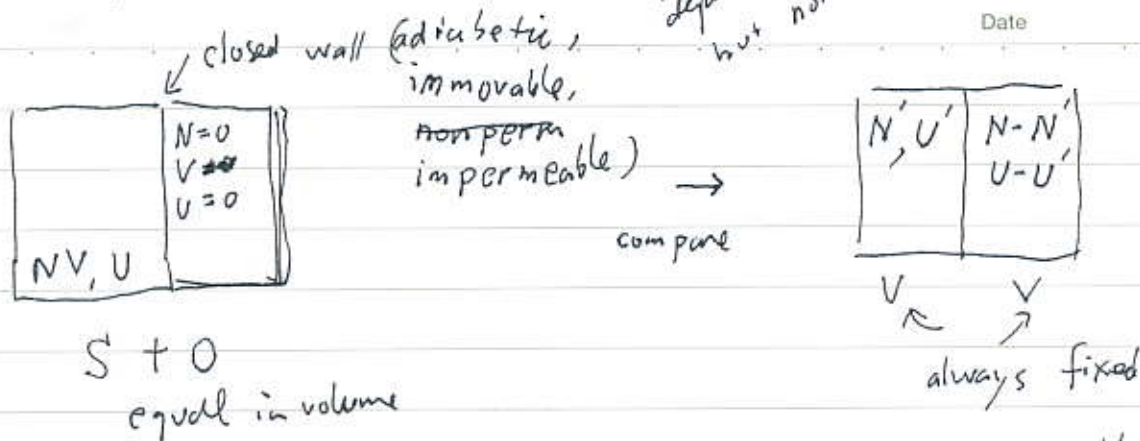
$$\frac{E(T) - E(0)}{T} = - \langle m \gamma v^2 \rangle + \langle R v \rangle = -P + 0$$

 $T \rightarrow \infty$

1

$$S = k_B N \ln(V U^{\frac{3}{2}}) + \text{const}$$

↑ depend on N
but not on V & U



what is $N' U'$ if partition is removed (becomes permeable)?

~~$$S_{\text{tot}} = k_B N' \ln(V U'^{\frac{3}{2}}) + k_B (N-N') \ln(V (U-U')^{\frac{3}{2}})$$~~

Need N -dependence

$$S = k_B N \ln(V U^{\frac{3}{2}}) + \frac{5}{2} N k_B (1 - \ln N) + \text{const}$$

$$S_{\text{tot}} = k_B N' \ln(V U'^{\frac{3}{2}}) + \frac{5}{2} N' k_B (1 - \ln N')$$

$$+ k_B (N-N') \ln(V (U-U')^{\frac{3}{2}}) + \frac{5}{2} (N-N') k_B (1 - \ln(N-N'))$$

$$= k_B (N-N') [\ln V + \ln (U-U')^{\frac{3}{2}}]$$

$$= k_B N' \ln V + k_B N' \ln U'^{\frac{3}{2}} + k_B (N-N') \ln V + \frac{k_B (N-N')}{2} \ln (U-U')^{\frac{3}{2}}$$

$$+ \frac{5}{2} N' k_B + \frac{5}{2} (N-N') k_B - \frac{5}{2} N' \ln N' - \frac{5}{2} k_B (N-N') \ln(N-N')$$

$$= k_B N \ln V + \frac{5}{2} N k_B + k_B N' \ln \frac{U'^{\frac{3}{2}}}{(U-U')^{\frac{3}{2}}} + k_B (N-N') \ln (U-U')^{\frac{3}{2}}$$

$$- \frac{5}{2} k_B [N' \ln N' + (N-N') \ln(N-N')]$$

$$\frac{\partial S_{\text{tot}}}{\partial N'} = 0 = k_B \ln U'^{\frac{3}{2}} - k_B \ln (U-U')^{\frac{3}{2}}$$

$$- \frac{5}{2} k_B [\ln N' + \ln(N-N')]$$

$$1 + \frac{U'^{\frac{3}{2}} (N-N')^{\frac{3}{2}}}{N'^{\frac{3}{2}} (U-U')^{\frac{3}{2}}} = 0$$

$$\left(\frac{N-N'}{N'} \right)^{\frac{3}{2}} \left(\frac{U'}{U-U'} \right)^{\frac{3}{2}} = 1$$

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$$\frac{\partial S_{tot}}{\partial U'} = \left[k_B N' \left[\frac{1}{U'^{\frac{3}{2}}} \right] + k_B (N-N') \frac{1}{(U-U')^{\frac{3}{2}}} (-1) \right] \frac{3}{2} = 0$$

$$\frac{N-N'}{N'} = \frac{(U-U')^{\frac{3}{2}}}{U'^{\frac{3}{2}}} \rightarrow$$

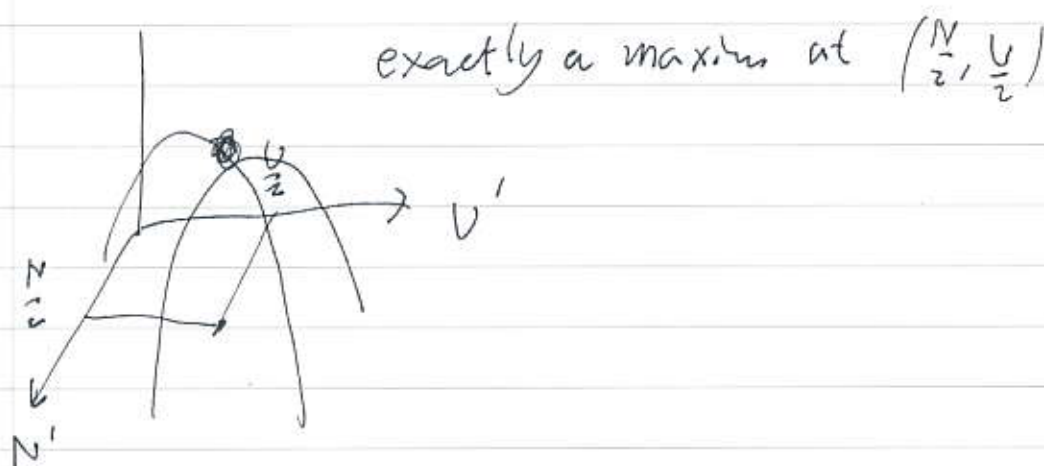
$$\left(\frac{N-N'}{N'} \right)^{\frac{3}{2}} \cdot \left(\frac{N-N'}{N'} \right) = 1 \quad \left(\frac{N-N'}{N'} \right)^{\frac{7}{2}} = 1$$

$$\frac{N-N'}{N'} = 1 \quad N-N' = N' \quad N = 2N'$$

$$N' = \frac{N}{2} \text{ as expected.}$$

$$\left(\frac{U-U'}{U'} \right)^{\frac{3}{2}} = 1 \quad U-U' = U' \rightarrow U' = \frac{U}{2}$$

i.e., ~~part~~ particles distributed evenly has max S'



$$dU = TdS - pdV + \mu dN$$

equilibrium statistical
density operator

 $\hat{\rho}$

$$\beta = \frac{1}{k_B T}$$

Date

□ canonical $\hat{\rho} \propto e^{-\beta \hat{H}}$ $\text{Tr}(\hat{\rho}) = 1$ fix the constant

$$dF = -SdT - pdV + \mu dN \quad Z = \text{Tr}(e^{-\beta \hat{H}}) \xrightarrow{\text{Helmholz free energy}} F = -k_B T \ln Z$$

$$\hat{\rho} = \frac{1}{Z} e^{-\beta \hat{H}} \quad \beta F = -\ln Z = \ln Z^{-1}$$

$$e^{\beta F} = Z^{-1}$$

$$= e^{-\beta(\hat{H} - F)}$$

compare $F = U - TS$

$$\text{Tr} \rightarrow \int \frac{dP \cdot dQ \dots}{h^{3N} N!}$$

$$\Psi \rightarrow e^{-\beta(U - F)} = e^{-\beta(TS)}$$

$$= e^{-\frac{S}{k_B}}$$

entropy

□ grand canonical

$$\hat{\rho} \propto e^{-\beta(\hat{H} - \mu \hat{N})}$$

$$Z = \text{Tr} e^{-\beta(\hat{H} - \mu \hat{N})}$$

Fock space = 0 particles $\otimes$ 1 particle $\otimes$ 2 particles $\otimes$...
Hilbert space Hilbert space Hilbert space ...

$$\hat{\rho} = \frac{1}{Z} e^{-\beta(\hat{H} - \mu \hat{N})}$$

$$= e^{-\beta(\hat{H} - \mu \hat{N} - \Psi)} \text{ again}$$

$$\text{Tr} \hat{\rho} = 1$$

$$\Psi = -k_B T \ln Z$$

$$= F - \mu N = U - TS - \mu N = -PV$$

$$e^{-\beta(U - \mu N - \Psi)}$$

$$= e^{-\beta(TS)} = e^{-\frac{S}{k_B}}$$

$$d\Psi = -SdT - pdV - Nd\mu$$

□ isothermal - isobaric ensemble (fix T, p, N)

$$G = F + pV = U - TS + pV = \text{Gibbs free energy} \quad e^{-\beta(TS)} = e^{-\beta(U + pV - G)}$$

$$dG = -SdT + Vdp + \mu dN$$

$$\hat{\rho} = e^{-\beta(\hat{H} + p\hat{V} - G)}$$

$$\text{Tr} \hat{\rho} = 1 \rightarrow G = -k_B T \ln \text{Tr}_{\hat{H}, \hat{V}} [e^{-\beta(\hat{H} + p\hat{V})}]$$

closed one
solve $-pV$

□ microcanonical

$$\hat{\rho} = \begin{cases} \sum |\varphi\rangle\langle\varphi| \cdot c & \leftarrow \text{sum over projector} \\ \varphi \in (E, U + \Delta) \\ 0 \end{cases}$$

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$$\text{Tr} |\varphi\rangle\langle\varphi| = 1$$

$$\text{Tr}(\hat{\rho}) = 1 = c \text{Tr} \sum_{\varphi} |\varphi\rangle\langle\varphi| = c \Omega \quad \begin{matrix} \text{state \# of microstates} \\ c = \Omega^{-1} \end{matrix}$$

=

$$\hat{\rho} = \frac{1}{\Omega} \sum_{\varphi} |\varphi\rangle\langle\varphi|$$

$$k_B \ln \Omega = S$$

$$\ln \Omega = S/k_B$$

$$= e^{-S/k_B} \sum_{\varphi} |\varphi\rangle\langle\varphi|$$

$$\Omega = e^{S/k_B}$$

↑ also in this for if we take $\sum_{\varphi} |\varphi\rangle\langle\varphi| \approx 1$